

Reference excerpt 02-025

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a during the $(i - 1)$ -th cycle, the parameters t_{xi} and t_{vi} are dened similarly. The parameters t_{ac} , t_{ax} , t_{xv} and t_{vy} represent the time interval between the waves a and c, c and x, x and v, v and y, respectively (see Fig. 2) and are measured by an operator over a selected cardiac cycle. The detection algorithm takes into account their possible modification during subsequent cycles, by means of the parameter (see Eq. (1)). Once identified the waves a_i , c_i , x_i , v_i e y_i on each cardiac cycle, the intervals t_{axi} and t_{vxi} are calculated for each cardiac cycle. The parameters dened in Eq. 1 are used to dene the two quantities $a_i = a_i - x_i$ $v_i = v_i - y_i$ (2). The average and the standard deviation for each parameter a_i , v_i , a_i , v_i , t_{axi} and t_{vxi} are calculated over several cardiac cycle. An illustrative JVP trace is presented in Fig.3 where the waves a, c, x, v and y are visible. The parameters a_i , x_i , v_i and y_i for each cardiac cycle i were identified by the algorithm. However, about 10 % of the points are normally not properly detected by the algorithm, while it was possible to identify them manually (see Fig. 4). The parameters a_i , v_i , t_{axi} and t_{vxi} were calculated as in Eq. (2). The mean values and standard deviations of the parameters are presented in Table 1. JVP and ECG time relationship. The time interval between the wave a and the events P and R